SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26041
- **Problem Statement Title-** AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector
- **Organization-** Department of Higher & Technical Education, Government of Jharkhand
- **Theme-** Smart Education (Vocational & Skill Training)
- **Sub-Theme-** Blockchain & Cybersecurity (Tamper-Proof Verification)
- **PS Category-** Software
- **Team ID-** 53476
- **Team Name (Registered on portal) :** SHADOW STACK



SHADOW STACK

THE SOLUTION

AI-POWERED SEMI OFFLINE AR SAFETY TRAINING & VERIFIABLE CERTIFICATION

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PROPOSED SOLUTION

- Android-first AR training running in mobile browsers without requiring ₹50,000 VR headsets.
- On-device AI evaluates worker safety decisions, PPE compliance, and procedure sequence.
- Preloaded, site-specific 3D training packages work 100% offline in underground mine pits.
- Tamper-proof digital certificates backed by NIST SHA-256 digests and Ethereum blockchain.

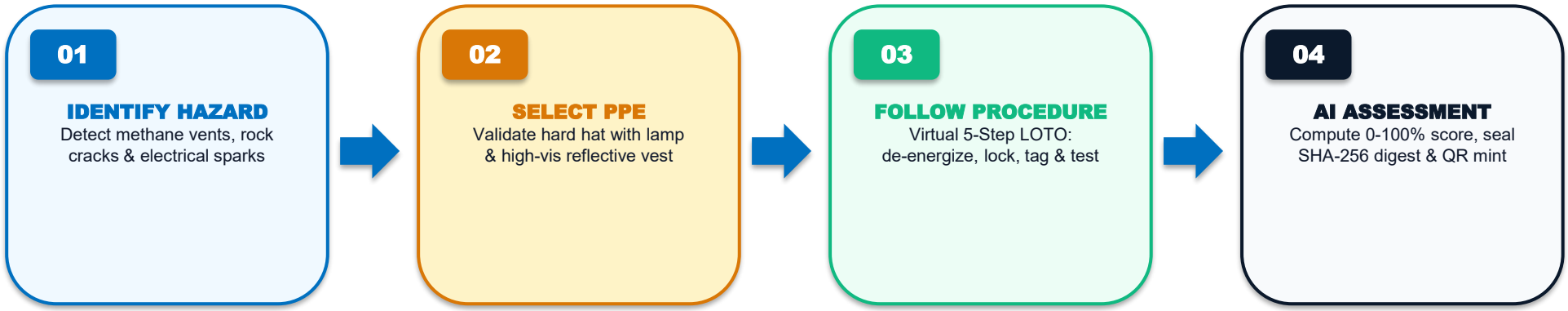
HOW IT ADDRESSES PROBLEM

- Replaces static, theoretical paper manuals with interactive spatial muscle-memory practice.
- Native Santali (sat) & Hindi (hi) localization bridges language barriers for tribal miners.
- Works seamlessly in remote deep coal shafts (Jharia, Dhanbad) with zero Wi-Fi or internet.
- Evaluates real operational competency and reaction time—not just passive attendance.

INNOVATION & UNIQUENESS

- AI + AR + edge inference: on-device 60 FPS WebGL simulation running on standard smartphones.
- Cryptographic .dat database: SHA-256 Merkle root integrity flags offline data tampering.
- Public QR verification portal: safety inspectors verify compliance at /verify without login.
- Configurable scenario schema: easily expands to gas leaks, roof falls, and machinery.

EXAMPLE FLOWCHART — MINING EMERGENCY & 5-STEP LOTO RESPONSE DRILL



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TECHNICAL APPROACH

Edge-first architecture: train offline, sync securely

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ADMIN / SITE
SETUP
Curriculum & Assets

→

TRAINING
PACKAGE
Preloaded 3D Model

→

ANDROID
APP / WEBAR
₹10k Phone Access

→

AR + AI
ENGINE
60 FPS Simulation

→

ASSESSMENT
+ LOCAL DB
SHA-256 .dat File

→

SECURE
SYNC
Ethereum Consensus

OFFLINE ZONE (UNDERGROUND DEEP MINE PIT)

- **Preloaded AR assets & site config:** Complete 3D colliery models cached locally on phone.
- **On-device AI state machine:** Evaluates procedure sequence and reaction timing without internet.
- **Encrypted local assessment/results:** Results stored in local .dat database sealed with SHA-256 digest.
- **Continuous training & scoring:** Workers practice and get graded autonomously in deep shafts.
- **Queued secure synchronization:** Pending records queue automatically for sync when network is detected.
- **Zero cloud dependence underground:** Mine site data is never continuously streamed to cloud servers.

VERIFIABLE CERTIFICATION

1. Certificate + Embedded QR
Contains worker ID, score & timestamp

2. NIST SHA-256 Hash Digest
Mathematical proof of tamper-free record

3. Ethereum Blockchain Anchor
Public on-chain verification at /verify

PRODUCTION TECHNOLOGY STACK & CRYPTOGRAPHIC STANDARDS

- **Frontend & WebAR Engine :** React 18, TypeScript, Three.js (WebGL 60 FPS hardware acceleration), Tailwind CSS, Vite, Lucide Icons
- **On-Device AI & Simulation :** State-Machine Procedural Validator, Spatial Raycasting, Gesture & Touch Orbit Controls, Vernacular Audio Engine
- **Security & Cryptography :** NIST FIPS 180-4 SHA-256 Hashing, Merkle Trees, Salted Key Derivation, Tamper-Evident .dat Snapshots
- **Ledger & Offline Persistence :** Ethereum / EVM Smart Contracts (Solidity), Ethers.js, Cloud Firestore Offline Cache, LocalStorage Edge DB

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FEASIBILITY & VIABILITY

Build a small, polished end-to-end prototype first

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FEASIBILITY

- **Mid-range Android target:** Optimized for Android 10+ smartphones without requiring external headsets.
- **WebAR zero-friction:** Runs directly in mobile browsers, eliminating app store installation barriers.
- **Preloaded site packages:** Executes 100% offline in underground mine shafts in under 10 seconds.
- **On-device AI inference:** Client-side evaluation preserves zero battery drain and high responsiveness.
- **Two complete safety modules:** Interactive Continuous Miner Drill and 5-Step Lockout/Tagout.
- **100/100 automated unit tests:** 12 test suites passing across authentication, crypto, and 3D engine.

CHALLENGES / RISKS

- **Low-light optical tracking:** Dark coal chambers and dusty air disrupt AR camera feature detection.
- **Tribal language & literacy:** Workers struggle with written English instructions and tech hesitation.
- **Offline data tampering:** Risk of tampering with locally stored safety training scores in deep pits.
- **Certificate privacy:** Worker PII must be protected while allowing public verification.
- **Device performance:** Thermal throttling and battery drain during intensive 3D rendering.

MITIGATION STRATEGIES

- **High-contrast WebGL virtual mode:** Dual-mode fallback with gyro and touch orbit controls requiring zero camera tracking.
- **Santali & Hindi voice HUD:** Multilingual audio guidance + large intuitive color-coded touch icons.
- **NIST SHA-256 Merkle sealing:** Local .dat files are cryptographically sealed; any modification triggers immediate tamper rejection.
- **Zero-knowledge hash verification:** Only cryptographic hashes stored on-chain; sensitive worker PII remains private.
- **Lightweight 60 FPS Three.js:** Low polygon models and optimized WebGL shaders guarantee 60 FPS without overheating.

IMPLEMENTATION PATH: PROTOTYPE → SCALED INDUSTRIAL DEPLOYMENT

1. AR Module 1 Mining Excavator
2. AI Assessment Sequence Logic
3. Security Layer NIST SHA-256
4. Blockchain Cert Ethereum Mint
5. Offline Sync .dat Database
6. Multi-Site Portal DGMS Dashboard

SHADOW STACK

IMPACT & BENEFITS

From one training module to a scalable industrial safety platform

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TARGET AUDIENCE & KEY BENEFITS

1. Target Audience :

- Mining & manufacturing workers: underground coal pits & metal smelting plants.
- Contract & newly inducted tribal laborers: direct training in Santali & Hindi.
- Safety trainers & compliance supervisors: automated tracking & instant grading.
- Government & institutional verifiers (DGMS): public zero-login verification.

2. Measurable Key Benefits :

- Higher engagement through practical AR: builds real spatial muscle memory.
- Measurable competency + audit trail: replaces easily faked paper attendance.
- Lower dependence on physical drills: saves ₹100s of crores in mock mine setup.
- Offline reach for remote worksites: operates where cloud tools fail completely.
- Tamper-evident certification: NIST SHA-256 + Ethereum public verification.

Feature	Manuals	VR Rigs	SafeAR
Hardware Cost	Zero (Paper)	₹50k+ VR	✅ **₹10k Phone**
Deep Mine Offline	✅ Yes (Passive)	❌ Needs PC	✅ **.dat DB**
Santali & Hindi	❌ English	❌ English	✅ **Synced**
Tamper-Proof	❌ Forged	❌ Local	✅ **SHA-256**
AI Competency	❌ None	⚠️ Basic	✅ **State AI**

SCALE: EXTENSIBLE MULTI-HAZARD SCENARIOS ACROSS MULTIPLE SITES & STATES

FIRE / EXPLOSION | **TOXIC GAS LEAK** | **CONFINED SPACE** | **HEAVY MACHINERY** | **MINE EVACUATION**

Deployable across major mining clusters: Jharia, Dhanbad, Bokaro, Ranchi, and Eastern Coalfields Ltd.

SHADOW STACK

RESEARCH & REFERENCES

Source basis and technical areas to verify before final SIH submission

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PROBLEM / POLICY BASIS

- **SIH 2026 Problem Statement 26041 :** Supplied problem statement by the Ministry.
- **Department of Higher & Technical Education, Government of Jharkhand :** Organization named in the PS.
- **PS Requirements :** Android 10+, WebAR modules, AI assessment, QR certificate verification, Hindi & Santali localization, offline operation, web admin dashboard.
- **PS Theme :** Smart Education (Vocational & Skill Training). Sub-theme: Blockchain & Cybersecurity.

TECHNICAL RESEARCH AREAS

- **Google ARCore & WebGL :** Mobile AR foundations & high-speed 3D rendering.
- **Android Developers :** Offline-first / local data synchronization patterns.
- **OWASP Mobile Top 10 :** Secure authentication, authorization & mobile security guidance.
- **NIST FIPS 180-4 :** Cryptographic hashing & tamper-evident Merkle digest standards.
- **Ethereum & EVM :** Transaction consensus, gas optimization & verifiable credential issuance.

LIVE EVALUATOR PROOF LINKS & ARTIFACTS

- **Live Production Application :** https://sihprototype-ten.vercel.app

(Three-Role Authentication: Worker, Trainer, Admin)

- **Interactive 3D Simulation :** https://sihprototype-ten.vercel.app/training/mining-basic-scene

(60 FPS WebGL Mining Excavator, LOTO Switch & Dust)

- **Public Blockchain Verifier :** https://sihprototype-ten.vercel.app/verify/SAFEAR-2026-DEMO01

(Zero-login public on-chain Ethereum consensus verification)

- **Automated Software Test Suite :** 100/100 Passed Unit Tests across 12 suites (Auth, Crypto, 3D Engine, Firebase Persistence)
- **Offline Database Inspector :** Accessible under Settings: live SHA-256 account inspection & .dat binary export/import